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# =====
# K-MEANS IMPLEMENTATION
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# Step 1: Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans

# -----
# Step 2: Create Dataset
# -----

data = {
    'Annual Income': [15,16,17,18,19,20,21,22,23,24,
                     60,62,63,65,67,68,70,72,74,75],

    'Spending Score': [39,40,41,42,43,44,45,46,47,48,
                      80,81,82,83,84,85,86,87,88,89]
}

dataset = pd.DataFrame(data)

print(dataset)

# -----
# Step 3: Select Features
# -----

X = dataset.iloc[:, :].values

# -----
# Step 4: Elbow Method
# -----

wcss = []

for i in range(1,11):

    kmeans = KMeans(
        n_clusters=i,
        init='k-means++',
        max_iter=300,
        n_init=10,
        random_state=0
    )

    kmeans.fit(X)

    wcss.append(kmeans.inertia_)

# -----
# Step 5: Plot Elbow Graph
# -----

plt.plot(range(1,11), wcss)

plt.title('The Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')

plt.show()

# -----
# Step 6: Train KMeans
# -----

kmeans = KMeans(
    n_clusters=2,
    init='k-means++',
    max_iter=300,
    n_init=10,
    random_state=0
)

y_kmeans = kmeans.fit_predict(X)

print("\nCluster Labels")
print(y_kmeans)

```

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print(y_kmeans)

# -----
# Step 7: Visualize Clusters
# -----

plt.scatter(
    X[y_kmeans == 0, 0],
    X[y_kmeans == 0, 1],
    s=100,
    c='red',
    label='Cluster 1'
)

plt.scatter(
    X[y_kmeans == 1, 0],
    X[y_kmeans == 1, 1],
    s=100,
    c='blue',
    label='Cluster 2'
)

# Plot Centroids
plt.scatter(
    kmeans.cluster_centers_[0,0],
    kmeans.cluster_centers_[0,1],
    s=300,
    c='yellow',
    label='Centroids'
)

plt.title('Clusters of Customers')

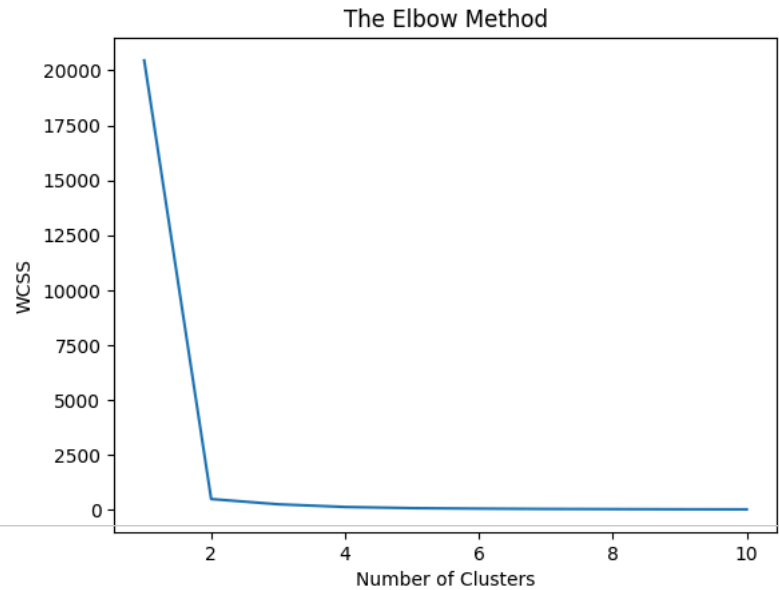
plt.xlabel('Annual Income')
plt.ylabel('Spending Score')

plt.legend()

plt.show()

print()
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	Annual Income	Spending Score
0	15	39
1	16	40
2	17	41
3	18	42
4	19	43
5	20	44
6	21	45
7	22	46
8	23	47
9	24	48
10	60	80
11	62	81
12	63	82
13	65	83
14	67	84
15	68	85
16	70	86
17	72	87
18	74	88
19	75	89



Cluster Labels
[1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0]

